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In [1]: import numpy as np

# Given points
points = {
    "P1": np.array([0.1, 0.6]),
    "P2": np.array([0.15, 0.71]),
    "P3": np.array([0.08, 0.9]),
    "P4": np.array([0.16, 0.85]),
    "P5": np.array([0.2, 0.3]),
    "P6": np.array([0.25, 0.5]),
    "P7": np.array([0.24, 0.1]),
    "P8": np.array([0.3, 0.2])
}

# Initial centroids
m1 = points["P1"]
m2 = points["P8"]

C1, C2 = [], []
cluster_labels = {}

# Assign points to nearest centroid
for p, val in points.items():
    d1 = np.linalg.norm(val - m1)
    d2 = np.linalg.norm(val - m2)
    if d1 < d2:
        C1.append(val)
        cluster_labels[p] = "C1"
    else:
        C2.append(val)
        cluster_labels[p] = "C2"

# Answers
print("P6 belongs to:", cluster_labels["P6"])
print("Population around m2:", len(C2))

# Updated centroids
new_m1 = np.mean(C1, axis=0)
new_m2 = np.mean(C2, axis=0)

print("Updated m1:", new_m1)
print("Updated m2:", new_m2)

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P6 belongs to: C1
Population around m2: 3
Updated m1: [0.148 0.712]
Updated m2: [0.24666667 0.2      ]

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In []: